## Business Recommendations & Operational Strategy

Introduction

This project focused on analysing hospital operational and financial data to generate insights that can support better healthcare management and decision-making.

The analysis was conducted using the provided healthcare dataset stored in MySQL and connected to Microsoft Power BI. SQL was used to inspect and prepare the data, while Power BI and DAX were used to develop key performance indicators, visualisations and an interactive dashboard.

The main objective was to understand hospital encounter activity, patient utilisation, encounter duration, procedure utilisation, healthcare costs and payer coverage.

The following evidence-based recommendations are established for hospital management.

Recommendation 1: Improve Operational Planning & Capacity Management

- **Finding Supporting Recommendation:** Encounter volume trends show substantial multi-year activity (peaking at 3,885 encounters in 2014 and 3,530 encounters in 2021). Ambulatory and outpatient visits consistently dominate operational volume (e.g., in 2021, ambulatory accounted for 36.91% and outpatient accounted for 40.17% of encounters).
- **Proposed Action:** Management should regularly monitor encounter volumes and class distributions to dynamically balance staffing and facility resources between routine outpatient clinics and emergency services.
- **Intended Benefit:** Supports proactive staffing adjustments, reduces patient wait times, and optimizes throughput in ambulatory clinics.
- **Limitations / Requirements:** Historical volume patterns reflect past utilization but may not fully predict acute surge events or public health emergencies.
- **Supporting Dashboard Evidence: Total Encounters KPI** (27,891) and **Encounter Distribution by Class Chart**.

Recommendation 2: Investigate and Monitor Extended Duration Encounters

- **Finding Supporting Recommendation:** Out of 27,891 total encounters, 27,816 visits (99.73%) were completed in under or equal to 24 hours, while only 75 encounters (0.27%) exceeded 24 hours.
- **Proposed Action:** Audit all 75 extended-stay encounters (>24 hours) to determine whether extended durations resulted from complex inpatient care protocols or operational bottlenecks in discharge planning.

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- **Intended Benefit:** Improves bed-turnaround efficiency, lowers non-reimbursable stay costs, and establishes operational benchmarks for long-stay care.
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- **Limitations / Requirements:** Electronic health records lack qualitative clinician progress notes to evaluate patient medical complexity directly.
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- **Supporting Dashboard Evidence: Average Encounter Duration KPI**, **Valid Duration Encounters KPI**, and **Encounters Over 24 Hours Analysis** (75 encounters / 0.27%).
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Recommendation 3: Strengthen Financial Monitoring & Claim Cost Oversight

- **Finding Supporting Recommendation:** Total claim costs vary significantly by payer. Medicaid holds the highest average total claim cost per encounter at **$6,205.22**, followed by Anthem at **$4,236.81**, Humana at **$3,269.30**, Blue Cross Blue Shield at **$3,245.58**, and Medicare at **$2,167.55**.
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- **Proposed Action:** Conduct quarterly claim audits across high-cost payer categories and align departmental budgets with actual claim reimbursement realizations.
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- **Intended Benefit:** Provides greater financial visibility, assists in insurance contract renegotiations, and improves financial forecasting accuracy.
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- **Limitations / Requirements:** Billed claim amounts represent gross charges rather than net collected revenue after negotiated payer discounts.
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- **Supporting Dashboard Evidence: Total Claim Costs KPI**, **Average Claim Cost KPI**, and **Total Claim Costs by Payer Chart**.
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Recommendation 4: Evaluate Zero-Coverage and Uninsured Encounters

- **Finding Supporting Recommendation:** A total of **13,586 encounters (48.71% of all visits)** had zero recorded payer coverage, with the NO_INSURANCE payer category averaging **$5,593.20** in total claim cost per encounter across 8,807 visits.
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- **Proposed Action:** Deploy financial screening counselors at registration to evaluate uninsured patients for Medicaid, Medicare Dual-Eligibility, or sliding-scale assistance prior to discharge.
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- **Intended Benefit:** Mitigates uncompensated care losses, reduces bad debt write-offs, and increases primary payer enrollment.
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- **Limitations / Requirements:** Requires integration with external state eligibility portals and real-time financial assessment tools.
- **Supporting Dashboard Evidence: Zero Coverage Encounters KPI** (13,586 encounters / 48.71%) and **Payer Coverage Analysis Chart**.